

# Blockchain and Application Stack

## L3 Data

PostgreSQL 16 (20  
entities, 3NF)  
Redis 7 · IPFS

## L4 API and Services

Express / Node 20 ·  
FastAPI  
WebSocket · EIP-  
712 · MCP  
middleware

## L5 Presentation

React 18 + Vite  
ERC-4337 AA ·  
MetaMask / email



## L1 Settlement / Network

Polygon zkEVM  
Cardona (ZK proofs)  
Ethereum Sepolia ·  
CCIP · The Graph ·  
Tenderly

## L2 Smart-Contract Platform

Solidity 0.8.20 ·  
OpenZeppelin v5  
UUPS · TimeLock ·  
RBAC